BECS 22623
Implementation of Numerical Methods using Appropriate software

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Outline

- Vector Calculation
- Matrix Calculation

Vectors

A vector is a one-dimensional array of numbers. MATLAB allows creating two types of vectors:

- Row vectors
- Column vectors

Row vectors are created by enclosing the set of elements in square brackets, using space or comma to delimit the elements.

$$r = [7 \quad 8 \quad 9 \quad 10 \quad 11]$$

Column Vectors are created by enclosing the set of elements in square brackets, using semicolon to delimit the elements.

$$c = [7; \quad 8; \quad 9; \quad 10; \quad 11]$$

Referencing the Elements of a vector

You can reference one or more of the elements of a vector in several ways. The ith component of a vector v is referred as $v(i)$. For example,

```
v = [1; 2; 3; 4; 5; 6]; % creating a column vector of 6 elements
v(3)
```

When you reference a vector with a colon, such as $v(:)$, all the components of the vector are listed.

```
v = [1; 2; 3; 4; 5; 6]; % creating a column vector of 6 elements
```

MATLAB allows you to select a range of elements from a vector. For example, let us create a row vector rv of 9 elements, then we will reference the elements 3 to 7 by writing $rv(3:7)$ and create a new vector named sub_rv.

```
rv = [1   2   3   4   5   6   7   8   9];
sub_rv=rv(3:7)
```

Vector operations

- Addition and Subtraction of vectors
- Scalar Multiplication of Vectors
- Transpose of a vector
- Appending Vectors
- Magnitude of a Vector
- Vector Dot Product
- Vectors with Uniformly Spaced Elements

Addition,Subtraction and Scalar Multiplication of vectors

Example

```
A=[7, 11, 15, 23, 9];
B=[2, 5, 13, 16, 20];
C=A+B;
D=A-B;
disp(C);
disp(D);
```

Example

```
v = [12   34   10   8];
m=5*v
```

Transpose of a Vector

The transpose operation changes a column vector into a row vector and vice versa. The transpose operation is represented by a single quote(').

Example

```
r = [1   2   3   4];
tr = r';  v = [1; 2; 3; 4];
tv = v';
disp(tr);   disp(tv);
```

Appending Vectors

MATLAB allows you to append vectors together to create new vectors. If you have two row vectors $r1$ and $r2$ with n and m number of elements, you write:

$$r = [r1, r2]$$

You can also create a matrix r by appending these two vectors, the vector $r2$, will be the second row of the matrix.

```
r = [r1; r2]
```

However, to do this, both the vectors should have same number of elements. Similarly, you can append two column vectors $c1$ and $c2$ with n and m number of elements. To create a column vector c of n plus m elements, by appending these vectors, you write:

```
c = [c1; c2]
```

You can also create a matrix c by appending these two vectors; the vector $c2$ will be the second column of the matrix.

```
c = [c1, c2]
```

However, to do this, both the vectors should have same number of elements.

Example

$r1 = [1\ 2\ 3\ 4];$
$r2 = [5\ 6\ 7\ 8];$
$r = [r1, r2]$
$rMat = [r1; r2]$
$c1 = [1;\ 2;\ 3;\ 4];$
$c2 = [5; 6; 7; 8];$
$c = [c1; c2]$
$cMat = [c1, c2]$

Magnitude of a Vector

Magnitude of a vector v with elements $v_1, v_2, ..., v_n$, is given by the equation:
$|v| = \sqrt{v_1^2 + v_2^2 + ... + v_n^2}$

- Take the product of the vector with itself, using **array multiplication** (.*). This produces a vector sv, whose elements are squares of the elements of vector v. $sv = v.*v$;
- Use the sum function to get the **sum** of squares of elements of vector v. This is also called the dot product of vector v. $dp = sum(sv)$;
- Use the **sqrt** function to get the square root of the sum which is also the magnitude of the vector v. $mag = sqrt(s)$;

Example

```
v = [1 :  2 :  20];
sv = v. * v; %the vector with elements
% as square of v's elements
dp = sum(sv); % sum of squares – the dot product
mag = sqrt(dp); %magnitude
disp('Magnitude:'); disp(mag);
```

Vector Dot Product

Dot product of two vectors $a = (a_1, a_2, .., a_n)$ and $b = (b_1, b_2, .., b_n)$ is given by $a \cdot b = \sum(a_i \cdot b_i)$ Dot product of two vectors a and b s calculated using the **dot** function.

$dot(a, b);$

Example

```
v1 = [2   3   4];
v2 = [1   2   3];
dp = dot(v1, v2);
disp('Dot Product:'); disp(dp);
```

Vectors with Uniformly Spaced Elements

MATLAB allows you to create a vector with uniformly spaced elements. To create a vector v with the first element f, last element l, and the difference between elements is any real number n, we write:

$$v = [f : n : l]$$

Example

```
v = [1 : 2 : 20];
sqv = v.^2;
disp(v); disp(sqv);
```

Matrix

A matrix is a two dimensional array of numbers. In MATLAB, you create a matrix by entering elements in each row as comma or space delimited numbers and using semicolons to mark the end of each row.

Example

$a = [1 \quad 2 \quad 3 \quad 4 \quad 5; 2 \quad 3 \quad 4 \quad 5 \quad 6; 3 \quad 4 \quad 5 \quad 6 \quad 7; 4 \quad 5 \quad 6 \quad 7 \quad 8;]$

Referencing the Elements of a Matrix To reference an element in the mth row and n^{th} column, of a matrix mx, we write

```
mx(m,n);
```

Example

$a = [1 \quad 2 \quad 3 \quad 4 \quad 5; 2 \quad 3 \quad 4 \quad 5 \quad 6; 3 \quad 4 \quad 5 \quad 6 \quad 7; 4 \quad 5 \quad 6 \quad 7 \quad 8;];$
$a(2, 5)$

To reference all the elements in the mth column we type $A(:, m)$.

Example

```
a = [1 2 3 4 5; 2 3 4 5 6; 3 4 5 6 7; 4 5 6 7 8;];
v = a(:, 4)
```

You also can select the elements in the mth through n^{th} columns, for this we write:

```
a(:,m:n)
```

Example

```
a = [1 2 3 4 5; 2 3 4 5 6; 3 4 5 6 7; 4 5 6 7 8;];
a(:, 2 : 3)
```

In the same way, you can create a sub-matrix taking a sub-part of a matrix.

```
a = [1 2 3 4 5; 2 3 4 5 6; 3 4 5 6 7; 4 5 6 7 8;];
a(:, 2 : 3)
```

Example

Let us create a sub-matrix *sa* taking the inner subpart of *a*:

```
a = [1 2 3 4 5; 2 3 4 5 6; 3 4 5 6 7; 4 5 6 7 8;];
sa = a(2 : 3, 2 : 4)
```

Deleting a Row or a Column in a Matrix

You can delete an entire row or column of a matrix by assigning an empty set of square braces [] to that row or column. Basically, [] denotes an empty array.

Example

```
a = [1 2 3 4 5; 2 3 4 5 6; 3 4 5 6 7; 4 5 6 7 8;];
a(4, :) = []
```

Next, let us delete the fifth column of *a*:

Example

```
a = [1 2 3 4 5; 2 3 4 5 6; 3 4 5 6 7; 4 5 6 7 8;];
a(:, 5) = []
```

Example

Let us create a 3 by 3- matrix m, then we will copy the second and third rows of this matrix twice to create a 4 by 3 matrix.

```
a = [1 2 3; 4 5 6; 7 8 9]
new_mat = a([2 : 3 , 2 : 3], :)
```

Matrix Operations

- Addition and Subtraction of Matrices
- Division of Matrices
- Scalar Operations of Matrices
- Transpose of a Matrix
- Concatenating Matrices
- Matrix Multiplication
- Determinant of a Matrix
- Inverse of a Matrix

Addition and Subtraction of Matrices

Example

$a = [1\ 2\ 3;\ 4\ 5\ 6;\ 7\ 8\ 9];$
$b = [7\ 5\ 6;\ 2\ 0\ 8;\ 5\ 7\ 1];$
$c = a + b$
$d = a - b$

Example

$a = [1\ 2\ 3;\ 4\ 5\ 6;\ 7\ 8\ 9];$
$b = [7\ 5\ 6;\ 2\ 0\ 8;\ 5\ 7\ 1];$

Scalar Operations of Matrices

When you add, subtract, multiply or divide a matrix by a number, this is called the **scalar operation**. Scalar operations produce a new matrix with same number of rows and columns with each element of the original matrix added to, subtracted from, multiplied by or divide by the number.

Example

```
a = [10 12 23; 14 8 6; 27 8 9];
b = 2;
c = a + b
d = a - b
e = a * b
f = a/b
```

Transpose of a Matrix

The transpose operation switches the rows and columns in a matrix. It is represented by a single quote(').

Example

$a = [10\ 12\ 23;\ 14\ 8\ 6;\ 27\ 8\ 9];$
$b = a'$

Concatenating Matrices

You can concatenate two matrices to create a larger matrix. The pair of square brackets'[]' is the concatenation operator. MATLAB allows two types of concatenations:

- Horizontal concatenation
- Vertical concatenation

When you concatenate two matrices by separating those using commas, they are just appended horizontally. It is used horizontal concatenation. Alternatively, if you concatenate two matrices by separating those using semicolons, they are appended vertically. It is called vertical concatenation.

Example

```
a = [10 12 23; 14 8 6; 27 8 9]
b = [12 31 45; 8 0 -9; 45 2 11]
c = [a, b]
d = [a; b]
```

Matrix Multiplication

- Consider two matrices A and B. If A is an $m \times n$ matrix and B is an $n \times p$ matrix, they could be multiplied together to produce an $m \times p$ matrix C. Matrix multiplication is possible only if the number of columns n in A is equal to the number of rows n in B.
- In matrix multiplication, the elements of the rows in the first matrix are multiplied with corresponding columns in the second matrix.
- Each element in the $(i, j)^{th}$ position, in he resulting matrix C, is the summation of the products of elements in i^{th} row of first matrix with the corresponding element in the jth column of the second matrix.
- Matrix multiplication in MATLAB is performed by using the $*$ operator.

Example

```
a = [1 2 3; 2 3 4; 1 2 5]
b = [2 1 3; 5 0 -2; 2 3 -1]
prod = a * b
```

Determinant of a Matrix

Determinant of a mateix is calculated using the **det** function of MATLAB. Determinant of a matrix A is given by $\det(A)$.

Example

$a = [1\ 2\ 3;\ 2\ 3\ 4;\ 1\ 2\ 5]$
$\det(A)$

Inverse of a Matrix

- The inverse of a Matrix A is denoted by A^{-1} such that the following relationship holds:

$$AA^{-1} = A^{-1}A = I$$

- The inverse of a matrix does not always exist. If the determinant of the matrix is zero, then the inverse does not exist and the matrix is singular.
- Inverse of a matrix in MATLAB is calculated using the **inv** function. Inverse of a matrix A is given by inv(A).

Example

```
a = [1 2 3; 2 3 4; 1 2 5]
inv(A)
```